



IN2901

**Software Development Project
Level 02 – Batch 24**

**ChainOps
IntelliCoders (05)**

Project Proposal

**Department of Information Technology
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Team Members

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1. Project Overview

ChainOps is a developer platform for teams that build and operate blockchain smart contracts. The platform brings the main stages of the smart-contract lifecycle into one controlled workflow: source-code integration, compilation and testing, security validation, approval, blockchain deployment, contract registration, runtime monitoring, alerts, notifications and audit history. The problem addressed is that these activities are often handled using separate scripts and tools, which can make it difficult to track which contract version is deployed to which network, whether validation passed, who approved a deployment, what transaction performed it and what important events occurred after release.

ChainOps will not build a new blockchain, Solidity compiler or security analyzer. Instead, it will integrate established tools such as GitHub, Hardhat/Foundry, Slither, OpenZeppelin and EVM-compatible test networks. The team's main contribution is the orchestration and operational-control layer: pipeline management, policies, approvals, deployment tracking, contract registry, event monitoring, notifications, real-time dashboards and the ChainOps AI Assistant.

2. Objectives

- Design and implement a fully functional smart-contract DevOps and operations platform.
- Provide authentication, organization/project management and role-based access for sensitive operations.
- Automate compilation, testing and security validation through a visible pipeline workflow.
- Provide controlled EVM smart-contract deployment with transaction and confirmation tracking.
- Maintain a versioned contract registry linking source commits, artifacts, networks, addresses and deployment history.
- Monitor deployed contracts, index relevant blockchain events and generate configurable alerts and notifications.
- Provide approval, governance and audit mechanisms for deployments and selected upgrade operations.
- Provide dashboards, reports and a read-oriented AI assistant for authorized operational queries

3. Project Scope

In Scope	Out of Scope
EVM-compatible Solidity projects; initially local EVM and selected public testnet(s).	Creating a new blockchain, consensus protocol, cryptocurrency or token economy.
GitHub repository integration and webhook-triggered/manual pipelines.	Creating a new Solidity compiler or vulnerability analyzer from scratch.
Project, environment, user/role, policy, approval, deployment and alert management.	Building a production-grade cross-chain bridge or supporting every blockchain network.
Build/test execution, security-analyzer integration and policy-based release gates.	Handling real user funds or requiring mainnet deployment for the academic demonstration.
Contract deployment, transaction status/confirmation tracking and contract registry.	Autonomous AI decisions or autonomous privileged blockchain transactions.
Runtime event monitoring, alert rules, in-app/email notifications and audit history.	Attempting to replace mature tools such as Hardhat, Slither or OpenZeppelin.
Real-time web dashboard and read-oriented ChainOps AI Assistant.	
Controlled upgrade workflow as an advanced feature if schedule permits.	

4. Key Features / Functionalities

Feature	What it Does
Authentication, Organization & Project Management	Users log in, work within an organization, manage projects/environments and receive role-based permissions.
Repository Integration	Connect a GitHub smart-contract repository and receive source-change information through webhooks.
CI/CD Pipeline	Run and visualize the release flow from compile and test through security validation, approval and deployment.
Build & Testing	Execute existing Solidity build/test tooling and collect results, artifacts and logs inside ChainOps.
Security Validation	Run an established analyzer, normalize findings and apply ChainOps release policies to allow, block or require review.
Approval & Governance	Create deployment/upgrade requests and allow authorized reviewers to approve or reject sensitive operations.

Deployment & Transaction Management	Initiate deployment to a configured EVM network and track transaction hash, status, confirmations and failures.
Contract Registry	Maintain versions, deployed addresses, networks, source commits, ABI/artifacts and deployment history.
Blockchain Monitoring	Listen for selected contract events and transaction activity and provide searchable runtime history.
Rules, Alerts & Notifications	Evaluate configured rules for failures/security findings/sensitive events and generate in-app or email notifications.
Audit Trail & Reporting	Record important actions and present deployment, approval, security and monitoring summaries.
ChainOps AI Assistant	Provide read-oriented operational Q&A such as deployment status, recent alerts, contract versions and pipeline failures using authorized ChainOps data.

End-to-end demonstration: A developer pushes a smart-contract change, ChainOps runs build/test/security stages, reviewers approve the release, a deployment worker submits it to an EVM testnet, the resulting contract address is stored, and monitoring detects a selected event that produces an alert.

5. Suggested Technologies

Area	Technology	Purpose
Frontend	React / Next.js + TypeScript	Developer portal, dashboards, forms, pipeline, security, deployment and monitoring interfaces.
Backend API	Node.js + TypeScript, NestJS (or equivalent)	Application APIs, orchestration, project/role management and service logic.
Platform Database	PostgreSQL	Users, organizations, projects, pipelines, contracts, deployments, findings, approvals, events, alerts and audit records.
Queue / Background Jobs	Redis + BullMQ (or equivalent)	Asynchronous build, security and deployment jobs, retries and worker coordination.
Smart Contracts	Solidity	Small demo contracts used to validate deployment, monitoring and upgrade workflows.
Smart-contract Framework	Hardhat or Foundry	Compilation, automated testing, artifacts and deployment scripts/tooling.
EVM Interaction	viem or ethers.js	RPC communication, transaction submission, ABI encoding/decoding and event access.
Security Analysis	Slither / established analyzer	Static-analysis findings used by the ChainOps security-validation stage.

Upgrade Tooling	OpenZeppelin Contracts / Upgrades	Upgradeable demo contracts and upgrade-safety validation for the advanced workflow.
Source Integration	Git + GitHub API / Webhooks	Repository connection, source revision metadata and pipeline triggers.
Real-time Communication	WebSockets / Socket.IO	Live pipeline, deployment, monitoring and notification updates.
Notifications	In-app notifications + SMTP/email provider	Deployment, approval, security and monitoring notifications.
Authentication / Access	Session/JWT-based authentication + RBAC	Identity and permissions for organizations, projects and sensitive operations.
Containerization	Docker / Docker Compose	Reproducible local development and isolated worker execution.
Blockchain Environment	Local EVM node + selected public EVM testnet(s)	Safe development and demonstration of real smart-contract deployments/events.
AI Assistant	LLM API with restricted read-only access to authorized ChainOps data	Operational Q&A without permitting the assistant to autonomously perform privileged blockchain actions.
Project Management / DevOps	Jira + GitHub + GitHub Actions	Backlog/sprints, issues, version control, pull requests, code review and CI/CD for the ChainOps platform itself.

6. Team Member Details

Member	Primary Ownership	Main Responsibilities
Member 1	Pipeline & Execution	Repository/webhook integration, pipeline execution, build/test flow, pipeline history and visible pipeline status.
Member 2	Blockchain Deployment & Contract Registry	Deployment requests/execution, transaction tracking, confirmations, contract versions, addresses and deployment history.
Member 3	Monitoring, Events & Alerts	Contract event monitoring, event history, monitoring dashboard, rules, alerts and runtime status.
Member 4	Security & Governance	Security findings, release policies, approval/rejection workflow, audit records and controlled upgrade flow.
Member 5	Platform, Access & AI Assistant	Authentication, organization/project/environment management, roles, shared dashboard/notifications and ChainOps AI Assistant.